Part 1: Hypotheses

Before running any experiments, predict the rejection rate for each defense configuration against each attack type. Write a percentage (0% or 100%) in each cell.

Experiment	Immediate Replay	Delayed Replay	Modified Replay
No defense	0%	0%	0%
Timestamp only	22-40%	70-100%	0-5%
Counter only	70%	50-80%	0-5%
All defenses	100%	100%	100%

Part 2: Experiment Results

Experiment 1: No Defense (Baseline)

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	5	0	0%
Delayed replay	5	0	0%
Modified replay	5	0	0%

Experiment 2: Timestamp Only

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	5	0	0%
Delayed replay	0	5	100%
Modified replay	5	6	0%

Experiment 3: Counter Only

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	0	5	100%
Delayed replay	0	5	100%
Modified replay	5	0	0%

Experiment 4: All Defenses

Attack Type	Accepted	Rejected	Rejection Rate
Immediate replay	0	5	100%
Delayed replay	0	5	100%
Modified replay	0	5	100%

Part 3: Summary Table

Compile all your results into a single table:

Defense	Immediate Replay	Delayed Replay	Modified Replay
No defense	0%	0%	0%
Timestamp only	0%	100%	0%
Counter only	100%	100%	0%
All defenses	100%	100%	100%

Part 4: Screenshots

Terminal output (experiment summary):

```
============================================================
EXPERIMENT SUMMARY
============================================================
Defense      Attack       Accepted   Rejected   Rate
----------------------------------------------------
none         immediate    5          0          0%
none         delayed      5          0          0%
none         modified     5          0          0%

[SAVED] Results saved to experiment_results.json
```

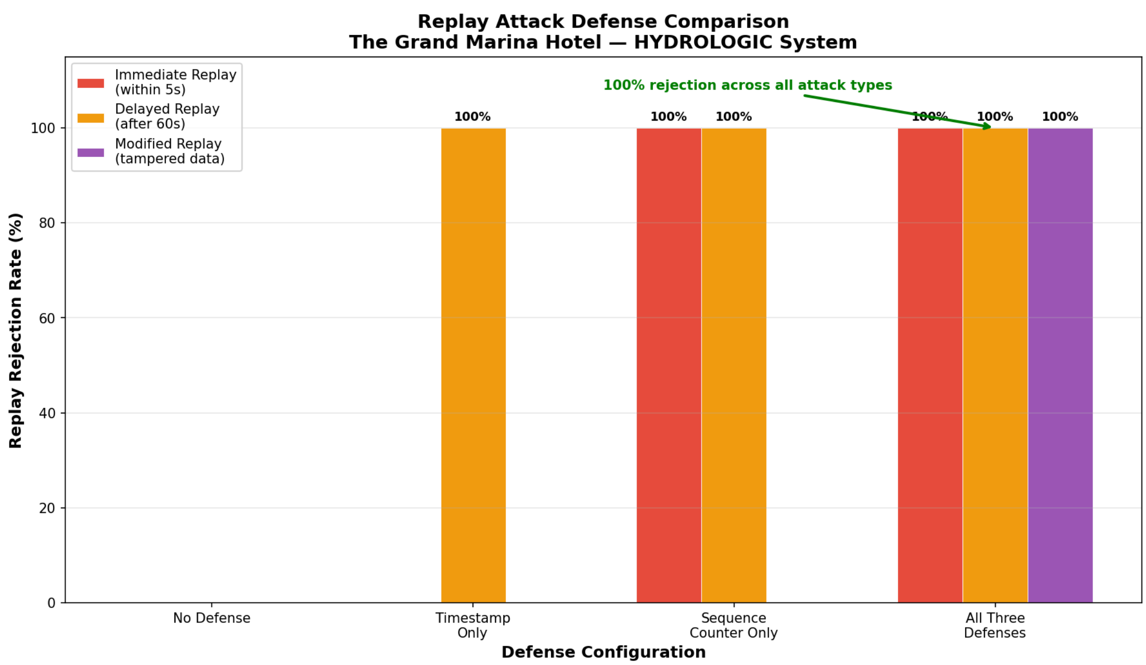
```
============================================================
EXPERIMENT SUMMARY
============================================================
Defense      Attack       Accepted   Rejected   Rate
----------------------------------------------------
timestamp    immediate    5          0          0%
timestamp    delayed      0          5          100%
timestamp    modified     5          0          0%

[SAVED] Results saved to experiment_results.json
```

```
==============================================================
EXPERIMENT SUMMARY
==============================================================
Defense       Attack        Accepted    Rejected    Rate
--------------------------------------------------------
counter       immediate     0           5           100%
counter       delayed       0           5           100%
counter       modified      5           0           0%
```

```
==============================================================
EXPERIMENT SUMMARY
==============================================================
Defense       Attack        Accepted    Rejected    Rate
--------------------------------------------------------
all           immediate     0           5           100%
all           delayed       0           5           100%
all           modified      0           5           100%
```

defense_comparison.png chart:



Part 5: Hypothesis Comparison

1. Which predictions were correct?

- My predictions for some of them were in a certain range so 70-100% I would say I got majority of them correct like counter test, and some of timestamp, and all of hmac.

2. Which results surprised you? Were there any attack/defense combinations you expected to block that didn't, or vice versa?

    - The outcome I was expecting was sort of predictable after reading through the documentation on how some of these defenses work and what they block, so I wasn’t too surprised.

3. Why did individual defenses have gaps? Think about what each defense checks and what it doesn't.

    - Some defenses have gaps due to what they are capable of doing, a timestamp defense cannot create unique signatures as its main goal is to track timing of the posted log data. This is why multiple defensive layers in any system is crucial because one defense cannot do everything.